

Question Number	Answer	Additional Guidance	Mark
4(a)	An answer that includes three of the following: • find the time for animal to start feeding (1) • find a suitable conditions for the animal to feed (1) • find a suitable method to for applying a touch/ suitable force (1) • find a suitable method of determining/measuring the extent of fan withdrawal (1)	Find/determine / identify eq Do not award method statements Accept find time interval between touches Accept measuring extension	Exp (3)

Question Number	Answer	Additional Guidance	Mark
4(b)	An answer that includes eight of the following: • clear statement of the dependent variable (1) • allow animal to acclimatise • method of touching the animal (1) • stated time intervals between touches or stated number of touches in a set time (1) • supply of organic particle to stimulate feeding activity (1) • identify one variable to be controlled and description of how it is controlled (1) • identify second variable to be controlled and description of how it is controlled (1) • repeat with another animal (1)	Accept record the length /extent of the withdrawal E.g. use of glass rod/cotton bud / touch with the same force MP6 and 7 e.g. temperature with thermostat in tank/waterbath Accept same age/sex	Exp (8)
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4(c)	• <u>raw</u> data table with headings and units, with means calculated from repeats (1) • scatter/line graph with labelled axes (1) • use of an appropriate (statistical) correlation test (1)	Accept any headers with appropriate units Do not accept description of calculating mean Do not accept units in the body of the table Accept bar graph and t test if comparing just two treatments	Exp (3)
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Question Number	Answer	Additional Guidance	Mark
4(d)	An answer that includes three of the following: • difficulty in determining extent of withdrawal (1) • difficult to determine same pressure/force of touch (1) • difficult to ensure each organism in the {same age /not already habituated} (1) • difficult to control concentration of organic particles (1)	Accept noise/vibration/water currents/light to shade	Exp (3)

(Total for question 4 = 17 marks)

